GEMINI 4 ARGON SF HACKATHON · OCTOBER 25, 2026

Virtual Lab

A lab of 28 Gemini agents that plans a study, runs it on real patient data, and has to beat Gemini 4 Argon answering alone.

github.com/SankarSubbayya/virtual_lab

One research question takes a real lab days

A REAL LAB

Days

Read the papers, plan the analysis, argue about it, write the code, run it on the data. Slow, but the numbers are measured.

AN AI CHATBOT

Seconds

Answers from memory. It never opens the data, so its numbers are educated guesses.

We wanted the speed of the chatbot **and** the rigour of the lab.

So we built the lab out of AI agents

1

Study

Learns the topic and writes cited notes the team keeps

2

Plan ×3

Three meetings at once: a lead scientist and three specialists

3

Critique

A critic attacks each plan until it holds up

4

Approve

A person approves the cost before anything runs

5

Run

The code runs on 1,980 real patients

6

Prove

A blind judge compares it with Argon alone

28 agents, each with one job. They pass work along like a relay, keep shared notes, and check each other’s work.

Data: METABRIC breast cancer cohort (Curtis et al., Nature 2012), public via cBioPortal

Argon leads; the others do the legwork

Role	Model	What it does
Lead scientist, critic, coder, writer, judge	**Gemini 4 Argon**	Runs the meetings, attacks the plans, writes the analysis code, writes the answer, judges
Specialists and the school	Gemini 3.8 Flash	Ideas from immunology, statistics and data; study notes
The lab bench	**Gemini Managed Agents**	Runs the code on 1,980 patients in a secure Google sandbox
First draft	Gemma 4	A quick answer from memory, checked later
The person in charge	You	Approves the budget; the approval becomes a hard spending cap

The data: a public breast cancer cohort

2,509

patients in the METABRIC study, from tumour banks in the UK and Canada, with long follow-up

1,980

of them have tumour gene-expression data, which the immune scores need: this is the lab's dataset

320

are triple-negative (ER, PR and HER2 negative), 168 of whom died during follow-up; the lab selects them itself

Public and de-identified. Downloaded by our code from the cBioPortal API (study **brca_metabric**): clinical details, survival, and 18 immune genes per patient.

Curtis et al., Nature 2012 · Pereira et al., Nature Communications 2016 · Rueda et al., Nature 2019

LIVE DEMO · ABOUT 2.5 MINUTES

“In triple-negative breast cancer, is more immune ‘killer’ activity in the tumour linked to longer survival, after accounting for age, stage and grade?”

Watch for: three meetings at once · the approval click · code running on real data · the blind verdict

The lab wins because it measures

8 of 9

blind tests won against Argon alone, judged twice with the order swapped (the lab must win both)

~20% lower

risk of death with more immune ‘killer’ activity: hazard ratio 0.76–0.80 in every run, measured on 227–320 real patients

2.5 min

from question to verdict, including the human approval (it took 11 minutes this morning)

$0.35

per question, about 120,000 tokens

Data: METABRIC breast cancer cohort, public via cBioPortal

Safe and fair by design

No medical advice

A question about one patient is turned into a research question about patients in general. Public data only.

A person holds the purse

Nothing runs on the data until someone approves the cost, and the approved budget is a hard cap.

A fair fight

The rival is Argon itself, with the same question and data description. The judge is blind, sees both orders, and the lab must win both.

What we built today

ALL OF THE CODE, WRITTEN TODAY

- The 28-agent relay on the Gemini Interactions API
- Analysis on Gemini Managed Agents, capped by the approved budget
- Fast mode: from 11 minutes to 2.5
- A live page showing every hand-off
- Safety guard, blind judge, 27 tests, two independent code reviews

github.com/SankarSubbayya/virtual_lab

BEFORE TODAY

An earlier prototype of the idea (Gemma and Claude). It never ran code on data and lost its blind test. We kept the design and rewrote everything.